

Generating Synthetic Anomaly Graph Network Dataset For Anti Money Laundering Prediction Using Generative Adversarial Network

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Abstract— In the domain of anti-money laundering (AML), obtaining sufficient labelled datasets to train anomaly detection models remains a significant challenge due to the sensitive nature of financial transactions. This study proposes a novel framework for synthetic anomaly graph data generation using Generative Adversarial Networks (GAN). The method encodes various graph topologies such as fan-in, fan-out, bipartite, feedforward, cycle, and scatter-gather into adjacency matrices to simulate complex transactional behaviours commonly associated with money laundering schemes. Through comparative evaluation against Variational Graph Autoencoder (VGAE) and traditional data augmentation techniques, the GAN-based model demonstrates superior capability in generating rich graph structures, reducing isolated nodes, and capturing diverse connectivity patterns. Despite higher reconstruction loss metrics (MSE and BCE), the GAN's ability to model intricate relational structures makes it well-suited for downstream graph-based anomaly detection tasks. The proposed approach addresses the scarcity of labelled financial graph data by offering a scalable, customizable method to generate synthetic datasets reflective of real-world AML patterns. These findings establish a foundation for further research in graph synthesis for financial crime detection using deep generative models.

Keywords—anti-money laundering, graph generation, synthetic dataset, anomaly detection, graph neural network, transactional pattern.

I. INTRODUCTION

Emergence of digital assets such as cryptocurrencies which is a revolution of the financial ecosystem. The advantage is low cost with peer-to-peer transfer [5]. Pseudonymity is the way that many cybercriminals, corruptors, terrorists and hackers have started to use cryptocurrencies[6][7]. Although cryptocurrencies are an application of blockchain technology, which implements 'Know Your Customer' (KYC) procedures to verify customers in the financial sector[8], money laundering transactions can still occur. Transaction patterns can be identified by unusual characteristics or anomalies without relying on identity. To prevent the damage caused by illegal transactions, many researches related to anomaly detection have been proposed[9]. However, accessing transaction datasets becomes very difficult due to privacy concerns [10].

Additionally, anomaly detection with conventional techniques cannot tackle the problem complexity of graph data [11] and has a problem with over-smoothing[12]. It is essential to make adjustments in methodology but the condition often faces the complexity in the algorithm, causing not easy to change. Therefore, the alternative solution is to define the anomaly network graph in an earlier process.

There are anomaly transactions which shape multiple difference patterns for anomaly detection in blockchain transaction networks [13]. Altman et al. [14] have proposed laundering patterns modelled to enhance the detection process of illicit financial transactions and published a realistic synthetic financial transaction dataset for anti-money laundering. It is an innovative solution based on problems obtaining the financial transaction dataset with facing privacy reasons [15]. Although real data were available, its labelling is inherently poor, causes the most laundering, and goes undetected.

Nevertheless, defining an anomaly network graph would be expected to change methodology from anomaly detection to pattern recognition. However, it is hard to identify an anomaly network graph where the anomaly has various forms and the graph may have a million connections containing between nodes and edges. For instance, let $G = (V, E)$ be a graph networks where $v_i \in V$ is a node with five number of degrees and E is a set of edges with v_0 has connected through five edges to one node. The node v_0 connects to v_1, v_2, v_3, v_4 and v_5 , then v_1 to v_5 and the network forms a star graph network and then v_1 walking to v_2 , v_2 walking to v_3 this is repeated until v_5 . Thus, each node has an interconnected link to others then it has formed a wheel graph network.

An adjacency matrix of a graph $G = (V, E)$ can represent graph containing a binary set, converting to vectors, and storing into variables. These vectors have been collected together with other various forms of the anomaly adjacency matrix. In this case, based on the pattern model proposed by Altman et al. [14], it contains various graph types such as fan-in, fan-out, bipartite, bipartite-feedforward, random, cycle, scatter-gather and gather-scatter. However, the matrix will be a training dataset but essentially each type of matrix has one combination. It would become problems, e.g., over-fitting[16], poor generalization[17], low-quality

embeddings[18], severe class imbalance[19] and difficult validation and evaluation[20], when it has a small dataset. Therefore, generating a combination of graph networks that describes anomalous transactions is a well-considered approach to tackle these issues and offer a solution.

Generative Adversarial Networks (GAN) is a machine learning model for generating dataset with pretending original dataset. It is based on the game, in the sense of game theory between two machine learning models, typically implemented using neural networks [21]. With consisting of two models a generator and a discriminator, GAN generates thousands of combination graph data which would be used as a synthetic dataset[22]. In addition, it can generate thousands of alternative graphs. Therefore, it could be a solution of the abovementioned challenging and this calculation can be obtained by using Equation (11).

To overcome the issues, the aim of this paper is to present a methodology for generating multiple combinations of graph networks using GAN. In the earlier step, the collection of various forms of anomaly adjacency matrix would be initialised as vector input for growing a number of combination anomalous transaction networks that can be used to predict anomaly finance transactions and to measure the performance of GAN, the construction loss function such as Mean Squared Error and Binary Cross Entropy, is used as shown at Equation (12) and Equation (13). In addition, the resulting anomaly patterns are also calculated.

The remaining of this paper is organized as follows. Section II describes data and methodology used in this study. Section III presents results and discussions. Section IV summarizes some conclusions and future studies.

II. DATA AND METHODOLOGY

A. Anomaly Graph Network Types

The form of anomalous transactions basically depends on suspicion arising from a series of financial transactions. It is possible that the defined form of anomalous transactions does not involve money laundering, such as financial transactions between a company and its subsidiary or a digital-based online business with a large number of customers. However, suspicion grew when each account within the transaction network conducted circular transactions and formed patterns, obscuring the fact that one account was transacting to withdraw the accumulated digital assets and transfer them to another account outside the transaction network that had formed a circular pattern. This condition also depends on our definition that there is no absolute type of transaction network but should we continue to allow digital asset transactions to constantly circulate from one account to another, thereby concealing the actual transactions, as if this were not an anomaly or there are massive transactions in a short period of time where digital assets are transferred from one account to another, which is not considered an anomaly, and if transactions can be carried out automatically, bypassing regulations and merely forming a transaction profile, is this considered normal? this is certainly a cause for concern for auditors.

Therefore, in this study, several types of anomaly transaction networks are defined, including fan-in, fan-out, bipartite, bipartite-feedforward, random, cycle, scatter-gather, and gather-scatter. Base on pattern, an adjacency matrix or neighbourhood matrix is formed. With the following equation

model. Then it is used as input of GAN as seen in Equation (11), and the result is a combination of adjacency matrices or membership matrices that are similar to the original data. This means that if it is converted back into a graph, the result is a graph with more degrees but with the same pattern, as shown at Fig. 1. The input data appears in the form of a star-shaped and cyclic graph, it is called a wheel graph, where each adjacent node connected by edges to one hub node, and these graphs forming into an adjacency matrix. So then, the adjacency matrix becomes the input matrix in GAN and produces variations form of adjacency matrices or wheel graphs with more hub degrees.

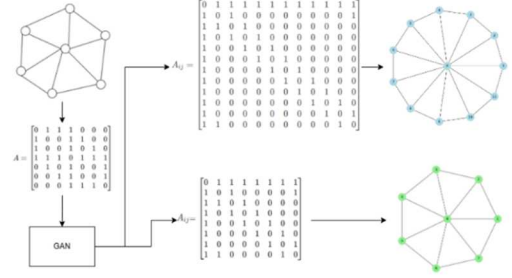


Fig. 1. Matrices schema input and output of GAN

Each anomalous graph network have modelled in mathematic. Let $G=(V, E)$ be a graph, where V is finite set of vertices and $v_i \in V$, $V = \{v_0, v_1, v_2, \dots, v_{n-1}\}$ and E is the set of edges that $E \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$. Base on graph function, typical definition found in each graph.

Definition 1 : Let $V = \{v_0, v_1, v_2, v_3, \dots, v_n\}$ be a set of external nodes, a cycle network graph E_{cycle} can be defined as

$$E_{cycle} = \{\{v_i, v_{i \bmod (n-1)+1}\} \mid v_i \in \{1, 2, \dots, n-1\}\} \quad (1)$$

where n is number of edges $v_{i \bmod (n-1)+1}$ is a pair node of v_i . To describe a hub that connected to other nodes shaping star pattern, as shown in Eq. (2).

Definition 2 : Star network graph E_{center} can be defined as

$$E_{center} = \{\{v_0, v_i\} \mid i \in \{1, 2, \dots, n-1\}\} \quad (2)$$

For the alternative transaction network that can be formed as input or output. Thus, the anomalous network form has transformed to wheel graph, as shown in Eq. (3).

$$E = E_{cycle} \cup E_{center} \quad (3)$$

Additionally, number of edges that own of wheel graph W_n is, that shown in Eq. (4).

$$|E| = (n-1) \text{ (from hub)} + (n-1) \text{ (from cycle)} = 2(n-1) \quad (4)$$

For the adjacency matrices, e.g., matrix A_{ij} that formed that shown in Eq. (5).

$$A_{i,j} = \begin{cases} 1 & \text{if } i = 0 \wedge 1 \leq j \leq n-1 \\ 1 & \text{if } j = 0 \wedge 1 \leq i \leq n-1 \\ 1 & \text{if } i, j \in \{1, \dots, n-1\} \wedge (|i-j| = 1 \vee |i-j| = n-2) \\ 0 & \text{else} \end{cases} \quad (5)$$

Where row and column 0 indicate connections from the hub to all outer nodes and vice versa. The connections to the

left and right neighbouring nodes in the cycle are $|i - j|$ in additions to that $|i - j| = n - 2$ is the connection between the first and last nodes or circular.

Definition 3 : Let V be a graph that $V = \{v_0, v_1, v_2, v_3, v_4, \dots, v_{n-1}\}$ with directed graph network from node v_i to v_j , E set of edges that E describes nodes connections set, suppose 8 number of vertices and set $E = \{(v_0, v_3), (v_0, v_4), (v_0, v_5), (v_0, v_6), (v_1, v_3), (v_1, v_4), (v_1, v_6), (v_1, v_7), (v_2, v_4), (v_2, v_5), (v_2, v_6), (v_2, v_8)\}$, a bipartite-feedforward network graph $E_{\text{bipartite-feedforward}}$, with adjacency function $f: V \rightarrow 2^V$ mapped a nodes to its target that adjacency node. Therefore, a node set if represented by notations with n is a number of nodes, k is a number of source nodes, m is a number of target nodes and $n = k + m$, as shown Eq. (6) below:

$$V = \{v_0, v_1, \dots, v_{n-1}\} = V_B \cup V_P \quad (6)$$

Where $V_B = \{v_0, v_1, \dots, v_{k-1}\}$ is a set of source nodes and $V_P = \{v_k, v_{k+1}, \dots, v_{n-1}\}$ is a set of target nodes. Thus, set of edges have directed only from source nodes to target nodes, as shown Eq. (7):

$$E \subseteq V_B \times V_P \quad (7)$$

Hence, a set of edges have formed as shown Eq. (8) below:

$$E = \{(v_i, v_j) \mid v_i \in V_B, v_j \in V_P, \text{ and } (v_i \rightarrow v_j) \text{ valid}\} \quad (8)$$

Then, adjacency function $f: V_B \rightarrow 2^{V_P}$ mapping each source nodes to a set of target nodes which are connected to it, as shown Eq. (9):

$$f(v_i) = \{v_j \in V_P \mid (v_i, v_j) \in E\} \quad (9)$$

If each source nodes connected to target nodes, then number of maximum edges is $|E|_{\max} = k \times m$. An adjacency matrix has notation $A \in \{0, 1\}^{n \times n}$ and element of adjacency matrix, as shown Eq. (10).

$$A_{ij} = \begin{cases} 1 & \text{if } v_i \in V_B \wedge v_j \in V_P \wedge (v_i \rightarrow v_j) \in E \\ 0 & \text{else} \end{cases} \quad (10)$$

For other graph patterns such as fan-in, fan-out, bipartite, random, scatter-gather, and gather-scatter, these patterns should be formed in adjacency matrix and then entered as GAN input. So that the overall output produced is $A_{h,i,j}$, which is the entire pattern generated. Where h is the pattern ID number sequence, i is the row, and j is the column.

B. GAN Architecture

A Generative Adversarial Network (GAN) consists of two main components, namely the Generator and the Discriminator, which work within a competitive framework[23]. The Generator is responsible for generating synthetic data from random noise vectors drawn from a specific distribution, such as a uniform or normal distribution. The goal of the Generator is to create data that closely resembles the original data distribution, making it difficult for the Discriminator to distinguish between the two. On the other hand, the Discriminator acts as a binary classifier that receives input in the form of original data and synthesized data, then provides a probabilistic assessment of the authenticity of the data. Both networks are trained simultaneously in a minimax

game, where the Generator seeks to maximize the Discriminator's error, while the Discriminator seeks to minimize its classification error[24].

Both networks were trained in a minimax game scenario, with the following objective function:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (11)$$

Through an iterative training process, both will continuously improve their performance until the Generator can generate data that closely resembles the original data.

The training process in Generative Adversarial Networks (GANs) takes place in iterative cycles involving two main stages: training the discriminator and training the generator. In each iteration, the discriminator is first trained using a combination of real data taken from the original data distribution and fake data generated by the generator. The goal of the Discriminator is to maximize accuracy in distinguishing between the two types of data, by assigning a label of 1 for real data and 0 for fake data. After that, the Generator is trained to produce synthetic data that can fool the Discriminator, by minimizing the classification error made by the Discriminator on the data produced by the Generator.

The network architecture in GAN is designed to adapt to the type and characteristics of the data to be generated. The generator functions to convert fixed-dimensional noise vector inputs into synthetic data that resembles real data, while the discriminator receives real data and synthesized results to evaluate their authenticity. Generator uses activation functions such as sigmoid or tanh to form the appropriate output format, while the Discriminator generally produces a single probability value between 0 and 1 as a representation of the authenticity of the input. The selection of the number of layers, neuron size, and regularization techniques such as dropout or batch normalization is very important to maintain training stability and prevent overfitting in both networks. A balanced architecture between the Generator and Discriminator supports more stable training and produces more realistic synthetic data. The resulting fake adjacency matrix will be used to check network patterns, such as in the image pattern matching process. Whether the network is the same as the network found in the pattern or not.

C. Reconstruction Loss Function

Reconstruction loss function is about measuring discrepancy between input and output are important process to produce best result as well as that control systems tools preventing incompatibility. The effectiveness of model that use generative proses such as Graph GAN, Variational Graph Auto Encoder (VGAE) and Augmentation Data can be measured using these measurements. Significance of choosing the reconstruction loss function in this case that modelling of generative graph, reconstruction loss helps ensure that the resulting graph resembles the original structure (e.g., in terms of adjacency matrix or feature matrix), rather than a meaningless random result. The two most prevalent reconstruction loss function used in generative process are Mean Squared Error (MSE) and Binary Cross-Entropy (BCE).

Mean Squared Error (MSE) is a loss function used to measure the magnitude of the error between predicted and actual values by calculating the average of the squares of their

differences. MSE works by taking each pair of predicted and actual values, calculating their difference, squaring the difference to ensure all values are positive and increase the penalty for large errors[25], then averaging them. The MSE value is always zero or greater, and the smaller the value, the more accurate the model's predictions are. MSE is commonly used in regression models, autoencoders, and graph generation to assess how well the model can reconstruct input data.

$$MSE = \frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (12)$$

Binary Cross Entropy (BCE) is a loss function used to measure prediction accuracy in binary classification problems by comparing the probability value of the prediction result (between 0 and 1) to the actual label (0 or 1)[26].

$$BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)] \quad (13)$$

BCE works by calculating the logarithmic difference between the prediction and the label: if the label is 1, then the loss is calculated by $\log(\hat{y})$ and if the label is 0, then it is used $\log(1 - \hat{y})$. Then, all loss values are averaged. The BCE value becomes larger if the prediction is far from the actual label, and smaller if the prediction is close to the correct value.

III. RESULT AND DISCUSSION

In this study, a number of adjacency matrices describing anomaly graphs were processed using GAN, and to compare the results, comparisons were made using several algorithms such as Variational Graph Auto Encoder (VGAE) and Augmentation Data (AD). The process involved checking several benchmarks based on comparisons of three algorithms, one of which was a pattern similar to the adjacency matrix against the results. However, before the pattern checking process is carried out, several checking performed, such as isomorphic, degree of node, symmetric and isolated nodes. In this case, the experiment was conducted on 1,200 graphs generated from the three algorithms, as shown below:

TABLE I. GRAPH CHECKING WITH SIZE 8X8 AND UNDIRECTED GRAPH

<i>Algorithms</i>	<i>Graphs</i>	<i>Mean of Degrees</i>	<i>Mean of Edges</i>	<i>Isolated Nodes</i>
GAN	1200	73.70	36.851	1399
VGAE	1200	56.7	28.35	77
Data Augmented	1200	31.468	15.734	35

In the GAN method, although it is capable of generating graphs with a high number of edges and degrees (an average of 36.85 edges and 73.70 degrees per graph), the number of isolated nodes is actually the highest, with 1,399 nodes out of a total of 1,200 graphs. This indicates that while many connections are created, their distribution is uneven, with some nodes having numerous connections while others remain completely unconnected. In contrast, VGAE generates graphs with more balanced connectivity: the average degree and number of edges are lower than in GAN, but the number

of isolated nodes is significantly fewer (77 nodes), demonstrating its ability to maintain a more balanced global structure. As for the Data Augmented approach, although it has the lowest average number of edges and degrees, its results are the most stable in terms of connectivity: only 35 nodes are isolated out of 1,200 graphs. This indicates that the method is more effective in keeping all nodes active even with minimal connectivity, making it highly useful for efficient and informative graph generation purposes.

Although the GAN method produces a high number of isolated nodes, GAN still has a number of important advantages in graph generation. GAN excels at mimicking complex and non-linear data distribution patterns, making it suitable for representing suspicious transaction patterns such as layering, fan-out, or circular routing, which are common in money laundering cases. In addition, GANs tend to generate graphs with high edge density, which is useful for simulating large-scale transaction scenarios. Their adversarial training process allows for high flexibility, including the possibility of integrating discriminators that are sensitive to specific motifs. This makes GANs ideal for use in graph data augmentation processes, especially when many samples are needed to train strong anomaly detection models that are resistant to pattern variations.

TABLE II. GRAPH CHECKING WITH SIZE 8X8 AND DIRECTED GRAPH

<i>Algorithms</i>	<i>Graphs</i>	<i>Mean of Degrees</i>	<i>Mean of Edges</i>	<i>Isolated Nodes</i>
GAN	1200	65.2333	32.6166	309
VGAE	1200	0	0	9600
AD	1200	34.368	17.184	8

The Table (2) shows a comparison between three graph generation methods, namely GAN, VGAE, and Data Augmented, based on the number of graphs generated, average degree, average number of edges, and number of isolated nodes. GAN generates graphs with complex structures and high connectivity, as evidenced by an average degree of 65.23 and an average number of edges of 32.61, with only 309 isolated nodes out of 1,200 graphs. In contrast, VGAE fails to establish connections between nodes, resulting in all nodes (9,600) being isolated, with average degree and edge values of zero. The Data Augmented method produces a better graph than VGAE with an average degree of 34.36 and only 8 isolated nodes, but it still lags behind GAN in terms of structural complexity. The main advantage of GAN lies in its ability to form realistic and connected graph structures, making it more effective for simulating complex patterns such as layering or smurfing in crypto-based money laundering transactions.

Isolated nodes have a negative impact because they eliminate contextual information, disrupt the learning process in graph-based models such as GNNs, and can cause bias in graph metric analysis. To address isolated nodes, pruning can be performed on irrelevant nodes, connections can be augmented by adding edges based on specific rules, focus can be placed on connected subgraphs, or special labelling can be applied if isolated nodes indicate anomalies. This handling is important for the model to effectively capture network patterns, especially in contexts like money laundering detection that rely on node relationships.

TABLE III. DIRECTED GRAPH PATTERN CHECK IN 1200 GRAPHS

Algorithm	Nodes Pattern Check					
	Cycle	Fan-in	Fan-out	Bipartite	Bipartite - Feedforward	Gather - Scatter
GAN	670	2240	2240	530	440	1870
VGAE ^a	0	1200	1200	0	1200	1200
AD	1180	1200	1200	0	0	1170

^a. Undirected Graph

The Table (3). shows the results of evaluating the structural patterns of 1,200 directed graphs generated by three methods, namely GAN, VGAE, and AD (Augmented Data), based on six types of patterns: cycle, fan-in, fan-out, bipartite, bipartite-feedforward, and gather-scatter. GAN significantly outperforms the others by generating more complex and realistic patterns, such as 670 cycles, 2,240 fan-in and fan-out patterns, 530 bipartite patterns, 440 bipartite-feedforward patterns, and 1,870 gather-scatter patterns, demonstrating its ability to form dynamic network structures. In contrast, VGAE, which generates undirected graphs, cannot form cycles or bipartite patterns, only recording minimal fan-in, fan-out, and feedforward patterns, while AD cannot form bipartite or feedforward patterns at all. This advantage demonstrates that GAN is more effective in constructing graphs that reflect complex interaction patterns, such as those required in money laundering simulations or financial network analysis.

TABLE IV. GRAPH LOSS CHECK MSE

Algorithms	Graph Loss Check MSE				
	200	400	600	800	1000
GAN	0.281	0.468	0.5	0.56	0.68
VGAE	0.43921	0.44233	0.43859	0.43898	0.4394
AD	0.326	0.327	0.327	0.327	0.327

Table (4) shows the Graph Loss values in the form of Mean Squared Error (MSE) from the 200th to the 1000th generation for three algorithms: GAN, VGAE, and AD. GAN has the lowest MSE at the beginning of training (0.281) but increases to 0.68 by the 1000th generation, while VGAE shows a stable but relatively high MSE value around 0.438, and AD has the most stable MSE at 0.327 without any increase during the training process.

Although GAN's MSE increases, this aligns with its ability to generate more complex and realistic graph structures, such as cycle patterns, fan-in, fan-out, bipartite, and a significantly smaller number of isolated nodes compared to other methods.

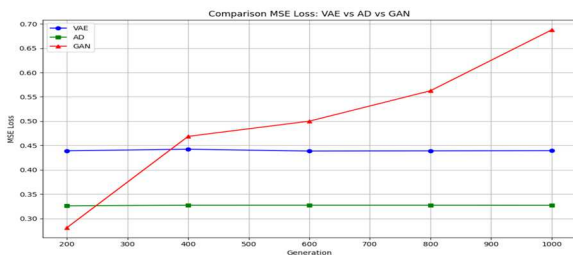


Fig. 2. Comparison of three algorithms in MSE. (MSE Loss VAE vs Data Augmentation vs GAN)

Conversely, VGAE and AD failed to form significant structural variations despite having more consistent MSE values.

TABLE V. GRAPH LOSS CHECK BCE

Algorithm	Graph Loss Check BCE				
	200	400	600	800	1000
GAN	28.125	46.875	50.0	56.25	68.75
VGAE	43.921	44.233	43.859	43.8982	43.9404
Data Augmented	6.005	6.027	6.028	6.027	6.025

Table (5) displays the Graph Loss values in the form of Binary Cross Entropy (BCE) from three algorithms—GAN, VGAE, and Data Augmented—at several training stages between generations 200 and 1000. GAN shows an increase in BCE values from 28.125 to 68.75, reflecting the process of exploring increasingly complex and varied graph structures as training progresses.

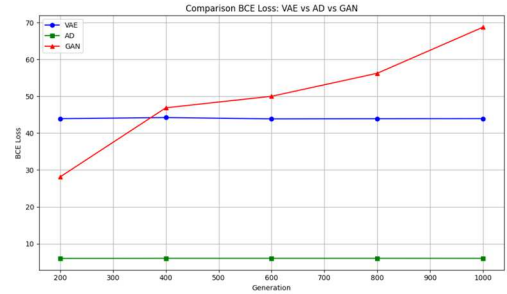


Fig. 3. Comparison of three algorithms in BCE. (BCE Loss VAE vs Data Augmentation vs GAN)

In contrast, VGAE has a relatively stable BCE value around 43.9 but fails to generate connected and realistic graph structures, while Data Augmented records a very low and constant BCE value (~6.02) but does not reflect generative learning, as it only replicates the initial data patterns without forming new structural variations. The advantage of GAN lies not in its low loss value but in its ability to generate graphs with rich patterns such as cycles, bipartite, feedforward, and gather-scatter, while minimizing isolated nodes. Thus, the advantage of GAN lies in the quality of the graph structures it generates, not merely in the error values.

IV. CONCLUSION

This study introduces a novel approach to generating synthetic anomaly graph network datasets for anti-money laundering (AML) prediction using Generative Adversarial Networks (GAN). By encoding graph patterns such as fan-in, fan-out, bipartite, feedforward, cycle, and scatter-gather structures into adjacency matrices, the proposed method enables the automatic generation of complex transactional structures that closely resemble real-world money laundering behaviour. Through comprehensive comparative experiments with VGAE and Data Augmentation techniques, GAN consistently demonstrates its superiority in producing structurally rich graphs with diverse connectivity patterns and fewer isolated nodes. Despite higher loss values in MSE and BCE at later training stages, GAN achieves better representational complexity, which is critical for modelling illicit financial activities.

The results validate GAN's capability to simulate varied transaction patterns that are difficult to obtain from real data due to privacy concerns. This approach not only provides an effective solution to the challenge of limited labelled datasets in AML but also paves the way for building robust graph-based anomaly detection models. Future work may explore hybrid GAN architectures and domain-specific constraints to further refine the quality of synthetic anomaly graph data and improve downstream detection performance.

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